[Aim: 100|100 in Maths]

आरंभ

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Lecture #2

#AIK²: $\boxed{\text{PLE2V}}$

$$a_1x+b_1y=c_1 \quad L_1$$
$$a_2x+b_2y=c_2 \quad L_2$$

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \rightsquigarrow$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Consistent

Inconsistent

#LP: If a pair of linear equations is consistent, then the lines will be:

(A) Parallel

(B) Always coincident

(C) Intersecting or coincident

(D) Always intersecting

atleast 1 sol^n

#LP: If $am \neq bl$, then the pair of equations $\boxed{ax + by = c}$, $\boxed{lx + my = n}$ has

(A) a unique solution

(B) no solution

(C) infinitely many solutions

(D) may or may not have a solution

$$\frac{a}{l} \neq \frac{b}{m} \qquad \frac{c}{n}$$

$$am \neq bl$$

$$\frac{a}{l} \neq \frac{b}{m}$$

i.e. $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \rightarrow$ intersecting

unique sol.

#LP : The pair of equations have $\frac{4}{x}+\frac{9}{y}=\frac{21}{xy}$ and $\frac{2}{x}+\frac{3}{y}=\frac{9}{xy}$

(A) a unique solution
(B) exactly two solutions
(C) infinitely many solutions
(D) no solution

$$\frac{4y+9x}{\cancel{xy}}=\frac{21}{\cancel{xy}}$$

$$4y+9x=21$$

$$\boxed{9x+4y=21}$$

$$\frac{2y+3x}{\cancel{xy}}=\frac{9}{\cancel{xy}}$$

$$2y+3x=9$$

$$\boxed{3x+2y=9}$$

$$\frac{a_1}{a_2}=\frac{9}{3}\Rightarrow 3$$

$$\frac{b_1}{b_2}=\frac{4}{2}\Rightarrow 2$$

$$\frac{a_1}{a_2}\neq\frac{b_1}{b_2}$$

intersecting

#LP: The value of t for which the pair of linear equations $\boxed{(t+3)x - 3y = t}$ and $\boxed{tx + ty + 12 = 0}$ have infinitely many solutions, is:

[CBSE Compartment, 2024]

(A) 6
(B) 0
(C) ~~-6~~
(D) 12

$$(t+3)x - 3y = t$$
$$tx + ty = -12$$

$\rightarrow \infty$ man sol $\Rightarrow$ coincident

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{t+3}{t} = \frac{-3}{t} = \frac{t}{-12}$$

$t \neq 0$

$$\frac{t+3}{\cancel{t}} = \frac{-3}{\cancel{t}}$$

$$t = -3-3$$

$$t = -6$$

$$\frac{-3}{t} = \frac{t}{-12}$$

$$36 = t^2$$

$$t = \sqrt{36} = \pm 6$$

~~6~~   -6

#LP:-The pair of equations $\boxed{3^{x+y}=81}$, $\boxed{81^{x-y}=3}$ has:

(A) No solution

(B) Infinitely many solution

(C) The solution $x=21/8, y=17/8$

(D) None of these

$$\boxed{(x^a)^b = x^{ab}}$$

$$3^{x+y} = 81$$

$$3^{x+y} = 3^4$$

on comparing

$$\boxed{x+y=4} \text{ — I}$$

$$81^{(x-y)} = 3$$

$$(3^4)^{x-y} = 3$$

$$3^{4(x-y)} = 3^1$$

comparing

$$4(x-y) = 1$$

$$\boxed{4x-4y=1} \text{ — II}$$

$$x+y=4$$

$$4x-4y=1$$

$$\frac{a_1}{a_2} = \frac{1}{4}$$

$$\frac{b_1}{b_2} = \frac{1}{-4}$$

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

intersecting

Unique sol

#LP: Two lines are given to be parallel. The equation of one of the lines is $\boxed{3x - 2y = 5}$. The equation of the second line can be (CBSE 2021)

(A) ~~$\overset{a_2}{9x} + \overset{b_2}{8y} = \overset{c_2}{7}$~~

(B) ~~$-12x - 8y = 7$~~

(C) $-12x + 8y = 7$ ✓

(D) $12x + 8y = 7$

$\parallel l \Longrightarrow$ $\boxed{\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} \neq \dfrac{c_1}{c_2}}$

$\boxed{\overset{a_1}{3x} - \overset{b_1}{2y} = \overset{c_1}{5}}$ — L_1

$$\frac{\cancel{3}}{-\cancel{12}\,4} = \frac{-\cancel{2}}{\cancel{8}\,4} \neq \frac{5}{7}$$

(C) $\dfrac{-1}{4} = \dfrac{-1}{4} \neq \dfrac{5}{7}$ ✓

#LP: Graphically, solve the following pair of equations:

$$2x + y = 6$$
$$2x - y + 2 = 0$$

Find the ratio of the areas of the two triangles formed by the lines representing these equations with the x-axis and the lines with the y-axis.

$\boxed{2x + y = 6}$ — I

x	0	3
y	6	0

(0, 6)
(3, 0)

$0 + y = 6$; $y = 6$

$\cancel{2}x + 0 = \cancel{6}\,3$; $x = 3$

$\boxed{2x - y = -2}$

x	0	-1
y	2	0

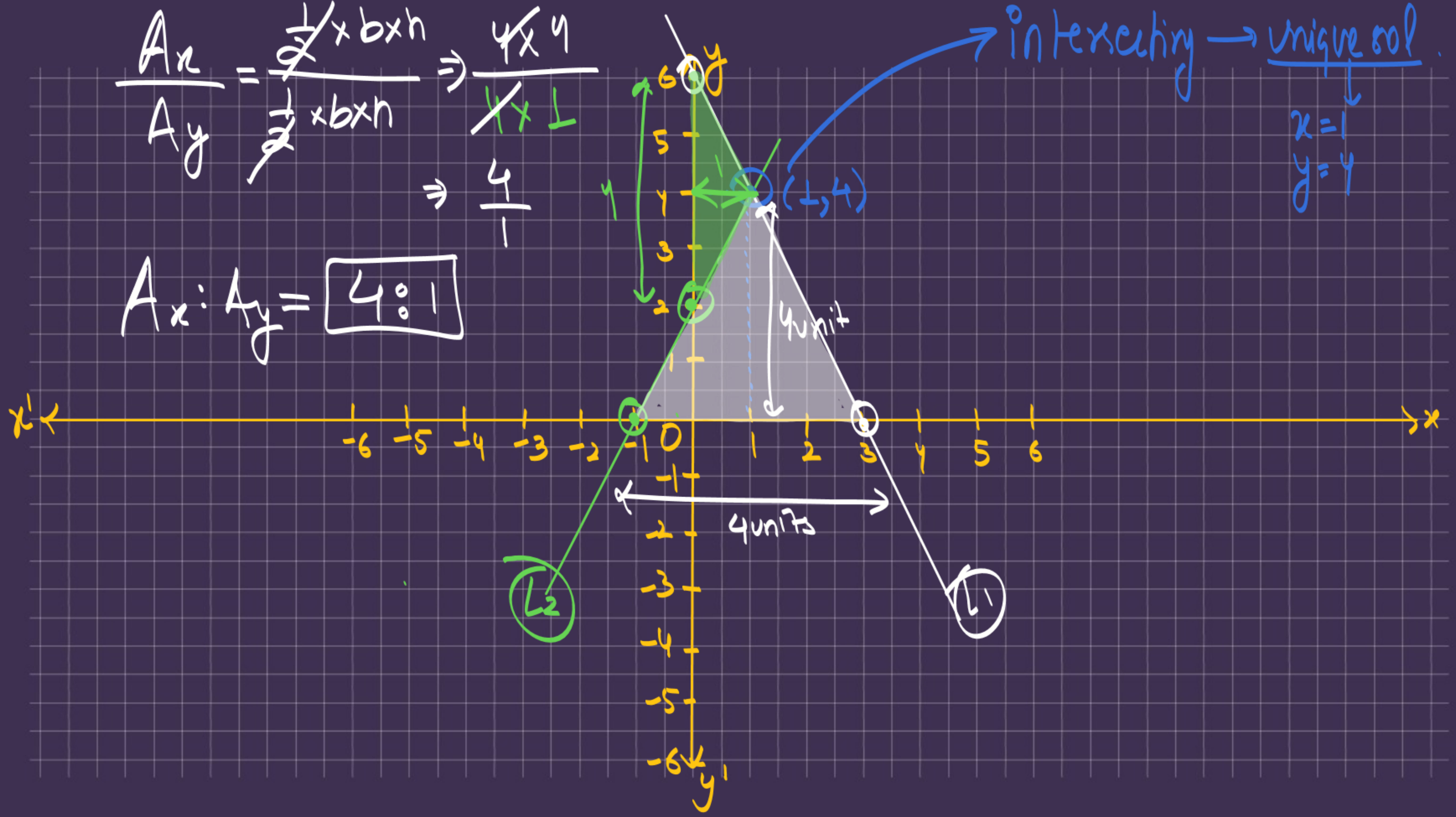
(0, 2)
(−1, 0)

$0 - y = -2$; $y = 2$

$\cancel{2}x - 0 = -\cancel{2}\,1$; $x = -1$

$$\frac{A_x}{A_y} = \frac{\cancel{\frac{1}{2}}\times b\times h}{\cancel{\frac{1}{2}}\times b\times h} \Rightarrow \frac{\cancel{4}\times 4}{\cancel{4}\times 1} \Rightarrow \frac{4}{1}$$

$$A_x : A_y = \boxed{4:1}$$

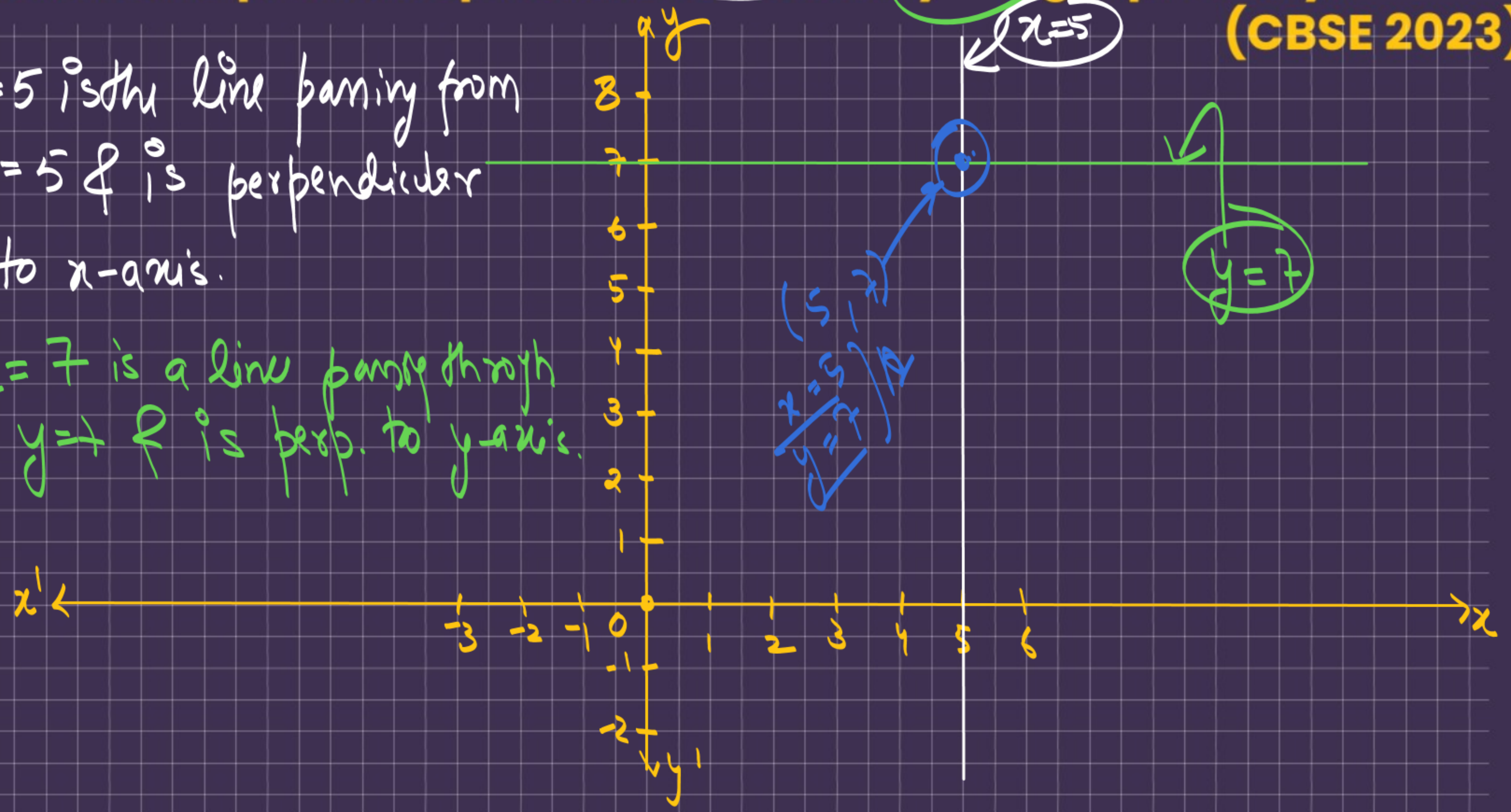


#LP: Solve the pair of equations x = 5 and y = 7 graphically.

(CBSE 2023)

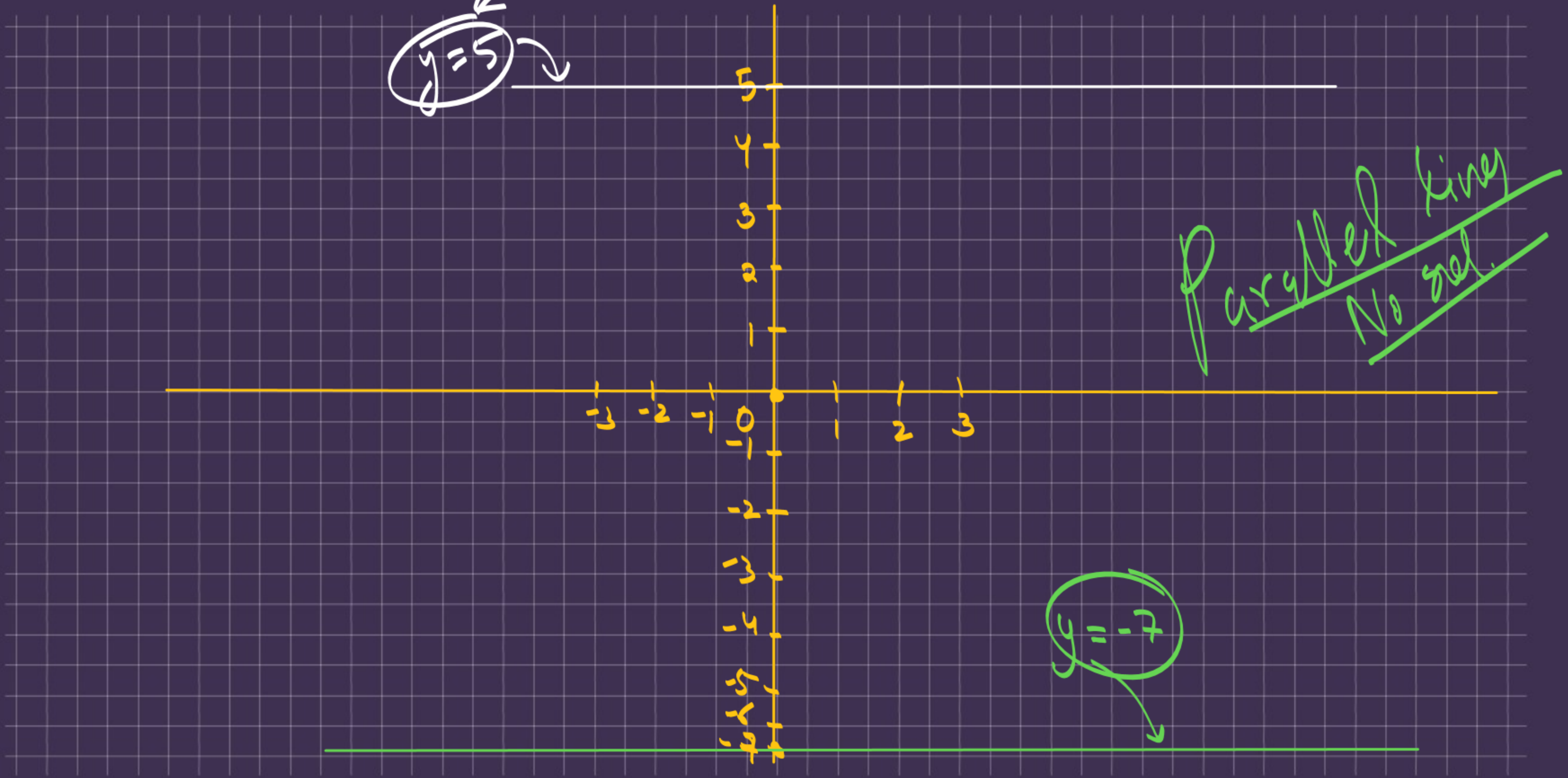
(a) $x=5$ is the line passing from $x=5$ & is perpendicular to x-axis.

(b) $y=7$ is a line passing through $y=7$ & is perp. to y-axis.



#LP: Graphically, solve the pair of linear equations:

$y - 5 = 0$ and $y + 7 = 0$ → $y = -7$



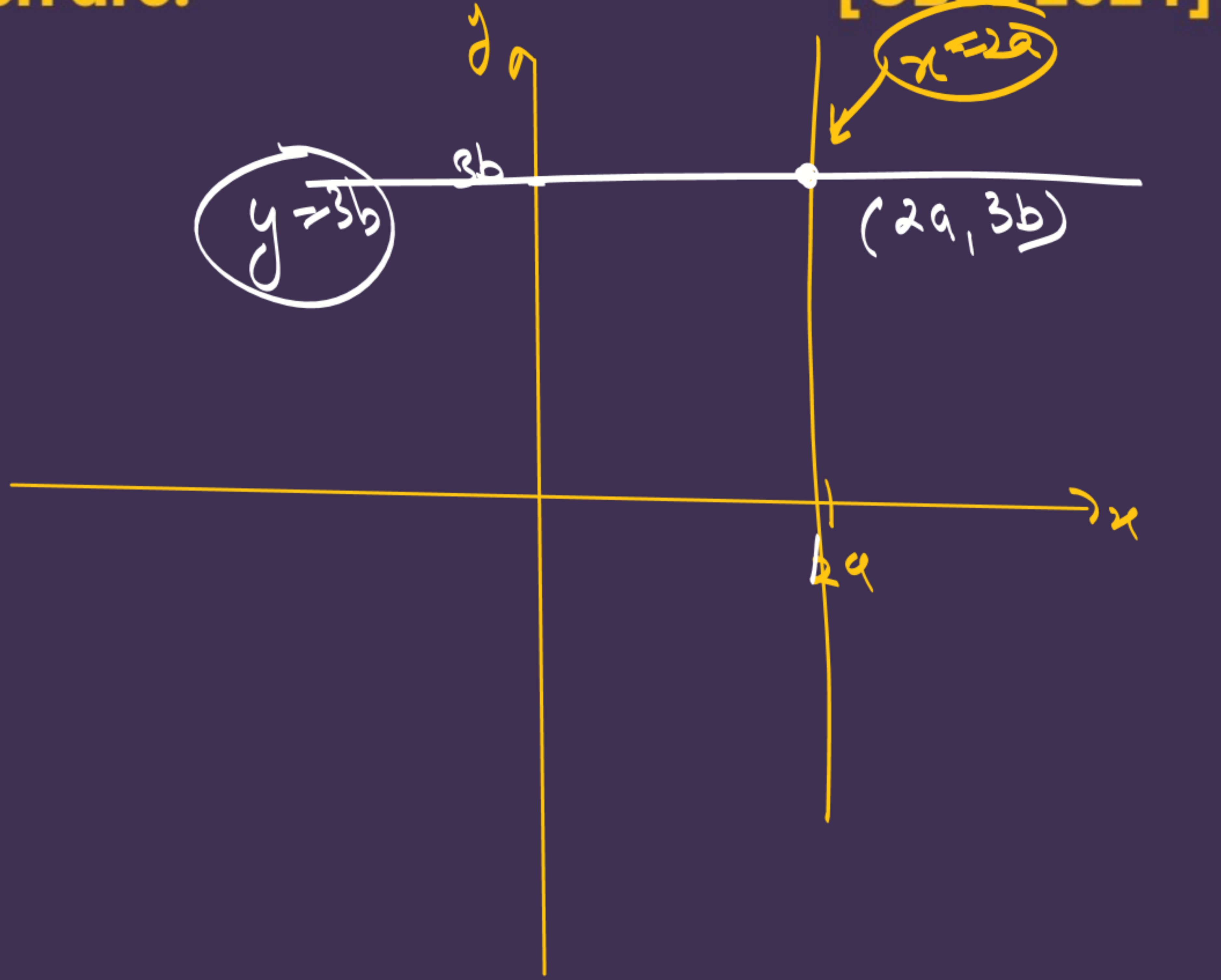
#LP: The pair of equations $x = 2a$ and $y = 3b$ $(a, b \neq 0)$ graphically represents straight lines which are: [CBSE 2024]

(A) coincident

(B) parallel

(C) intersecting at $(2a, 3b)$

(D) intersecting at $(3b, 2a)$



Next Toppers

h/w → next dpp

THANK YOU COODIESS !!